The Coercion Variable: On the Boundary Between What the Sovereign Must Do and What the Bond Can Do

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Abstract

The dominant tradition of legal philosophy from Austin [1832] and Hobbes [1651] through Weber [1919] and Hart [1961] has treated law as a system of texts paired with a coercive enforcement apparatus. The texts specify obligations; the apparatus — police, courts, sheriffs, ultimately the sovereign's monopoly on legitimate violence — ensures the obligations are met or punishes their breach. The exact articulation of the relationship between text and coercion varies across the tradition (command theory, secondary rules, legitimate violence, and so on), but the structural pairing is shared.

This paper observes that the bonded settlement primitive examined in the corpus's companion mechanism, institutional-economics, and political-economy papers performs the enforcement function of law in bilateral commerce *without* a coercive apparatus in the tradition's sense. There is no third party who applies force upon breach. There is no court that compels performance. There is no sovereign whose monopoly on violence stands behind the arrangement. The enforcement is performed by collateral that the parties themselves lock at the moment of consent, and that the protocol releases or forfeits according to the parties' subsequent behavior. The mechanism is, in the precise language of Elster [1979], a *pre-commitment* structure: self-binding at the moment of consent, with no external party in the position to apply or withhold force after the fact.

Pre-commitment via bond is qualitatively different from retributive coercion via the sovereign. The difference is not a matter of degree — a softer or kinder enforcement — but of mechanism: the relationship between obligation and enforcement runs through the parties' own resources and consent, not through an external authority's standing capacity to compel. Where the mechanism applies, it substitutes for the coercive apparatus. Where it does not apply — against criminal harm, against externalities affecting non-parties, against the provision of public goods, against status and rights adjudication — the coercive apparatus continues to do work that the bonded mechanism is structurally unable to do.

The substantive claim of this paper is therefore not that the sovereign is unnecessary or that the state should be reduced. It is that the political-philosophical question of legitimate coercion — which thinkers from Hobbes to Schmitt have addressed as if it ranged over the entirety of social cooperation — becomes a bounded question once a substantial part of bilateral commerce can be coordinated by pre-commitment rather than by sovereign-backed coercion. The state retains its load-bearing functions; the question of how it should exercise them retains its political-philosophical urgency; what changes is the *domain* over which the question must be answered. This relocation is the political-philosophical content of the substrate change, and it is the subject of this paper.

Keywords: coercion, sovereignty, monopoly on violence, pre-commitment, bonded enforcement, political philosophy, Hobbes, Weber, Hart, boundary of state authority

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1 Introduction

This paper sits at the intersection of political philosophy and the analysis of a particular coordination primitive. Its starting point is an observation that emerged in conversation about how to characterize the bonded settlement primitive relative to the dominant Western tradition on the nature of law. The observation is sharp enough to warrant explicit treatment.

The dominant tradition reads law as the pairing of texts (the obligations) with a coercive apparatus (the enforcement). The coercive apparatus performs two functions: it compels performance of obligation when a party would not otherwise perform, and it punishes breach as a deterrent against future breach by the same party or by others. The apparatus is grounded in the sovereign's monopoly on legitimate violence within a territory (Weber 1919). When the obligation is honored, the apparatus does no observable work; when the obligation is breached, the apparatus is invoked. The apparatus's standing capacity is what makes the texts effective even in cases where it is not invoked: the threat of enforcement does the work even when the actual enforcement is not required.

The bonded settlement primitive examined in the corpus's companion mechanism paper performs the enforcement function for bilateral commercial obligations *without* a coercive apparatus in the tradition's sense. The bond is locked at the moment of consent, by the parties themselves, from their own resources. The protocol's resolution mechanic releases the bond to the cooperative party and forfeits it from the defecting party, automatically and without discretion. There is no sovereign behind the arrangement, no court to which a party can appeal for force, no agent in the position to apply or withhold enforcement after the fact. The enforcement is internal to the arrangement; the only external requirement is the public infrastructure on which the protocol runs.

This is not a marginal substitution. In bilateral commercial exchange, the coercive apparatus has historically been the guarantor of last resort against which all other enforcement mechanisms (reputation, repeated interaction, social pressure) operate. Removing the requirement for that guarantor changes the structural role those secondary mechanisms play. More fundamentally, it changes what kind of question the philosophical analysis of coercion is. We argue that this change is the substantive political-philosophical content of the substrate change.

The thesis. The thesis is structural and bounded. (1) Coercion is a substrate variable: in the dominant tradition, it is structurally load-bearing in the enforcement of bilateral commercial obligations. (2) The bonded primitive removes coercion as a substrate variable in this domain by substituting pre-commitment for retribution. (3) The substitution does not extend to domains in which the coercion was performing functions the bond cannot perform: criminal harm, third-party externality, public-goods provision, status, rights adjudication. (4) Where the substitution applies, the political-philosophical question of how to legitimate coercive power becomes irrelevant rather than corrected — the coercion is no longer being exercised, so the question of its legitimacy does not arise. (5) Where the substitution does not apply, the political-philosophical question retains its full urgency, now operating over a bounded rather than an unbounded domain.

What this paper is and is not. This paper is a short essay in the political philosophy of coercion. It does not extend the underlying mechanism, the institutional argument, or the political-economy diagnostic developed in the companion papers. It takes those contributions as given and asks what their joint implication is for the philosophical analysis of coercion that has organized political thought from Hobbes to the present.

The paper takes no position on whether the state should be larger, smaller, more interventionist, less interventionist, democratic, republican, or any other configuration. Those questions are real, and they survive the substrate change in the domain where coercion remains load-bearing.

The paper's claim is that the substrate change shrinks that domain by removing bilateral commerce from it, and that this shrinkage is the political-philosophical content of the change. The shrinkage is neither a libertarian victory nor a statist defeat — it is a relocation of where the question of legitimate coercion must be answered.

Position relative to the corpus. This paper is part of the law-and-civil-authority cluster (with the companion labor-law paper on subordination as legal axis and the companion displaced-populations paper on civil-status preconditions). It addresses what those papers presuppose: the coercive apparatus that historically stands behind both labor contracts and civil status, and what the substrate change implies for that apparatus when its commercial-enforcement function is substituted by bond. The political-economy trilogy in the B-series addresses adjacent questions (institutional-economic mechanism in B1, hegemonic visibility in B2, conditionality of the left/right partition in B3). The present paper addresses the political-philosophy question that the B-series brackets: what is removed when coercion is removed, and what continues to require it.

2 Coercion as the Third Substrate Variable

The companion partition-conditionality paper identified two substrate variables — subordination and coordination rent — on which the disagreement between the cooperative-Marxian and classical-liberal traditions had been organized. The bonded primitive removes both as load-bearing variables in bilateral commerce; the disagreement does not vanish but relocates from the substrate to the graph. The present paper identifies coercion as a third such variable, structurally adjacent to the first two and analyzed by a different tradition.

Coercion in the dominant tradition. Austin [1832] formulated the command theory of law: laws are commands of a sovereign, backed by sanctions for non-compliance. The pairing of command with sanction is constitutive of law on this view; an obligation without an enforcement mechanism is, in Austin's terms, a moral injunction rather than a legal one. Hart [1961] refined the analysis substantially, distinguishing primary rules of obligation from secondary rules of recognition, change, and adjudication, and arguing that not all law operates by sanction. But Hart retained coercive enforcement as one component of legal systems and acknowledged that, in practice, the enforceability of legal rules depends on the standing capacity of officials to apply force when required. Hobbes [1651] provided the upstream political-philosophical justification for the standing capacity: covenants without the sword are but words, and of no strength to secure a man at all.

Weber's distinction between *Macht* and *Herrschaft*. Weber [1922] sharpens the picture in a way the present argument relies on heavily. *Macht* is the bare capacity to bring about an outcome despite resistance — power in the most general sense. *Herrschaft*, often translated as "domination" or "authority," is a narrower category: *Macht* that is exercised through a stable relationship of command and obedience, in which one party has a recognized right to issue directives and another party a recognized obligation to comply. Weber's three pure types of legitimate *Herrschaft* — traditional, charismatic, and rational-legal [Weber, 1919] — describe how command-relationships are stabilized as authority. The Hobbesian sword Austin and Hart later analyze is *Herrschaft*-shaped: it presupposes the sovereign's recognized authority to exercise the coercive capacity, not just the bare capacity itself.

The bonded primitive does coercion-like work — the breaching party's bond is forfeit despite their preference — but it does that work in the *Macht* register without reaching for *Herrschaft*. There is no party whose recognized authority the forfeiture depends on; the bond is released or forfeit by public algorithm whose execution the parties have themselves authorized at the moment of consent. The primitive supplies the forcing function (*Macht*) without supplying or requiring the authority structure (*Herrschaft*) that the dominant tradition has assumed must accompany it. This is what makes coercion a variable: separating *Macht* from *Herrschaft* has been historically difficult precisely because the available enforcement architectures bundled them together; the bonded primitive is one of the first arrangements in which the bundle can be unbundled at the level of bilateral commerce.

The functional locus of coercion across these accounts is the same: it is what makes contractual obligation effective in cases where a party would not otherwise perform. Without it, parties who can defect at lower cost than they can perform will defect, and the system of obligation collapses into a set of pious recommendations. With it, performance becomes the rational strategy because the cost of breach exceeds the cost of compliance.

Coercion in commercial law specifically. Commercial law inherits the dominant tradition's structure. Contract enforcement runs through the courts; specific performance and damages awards rely on the state's capacity to compel transfer of property or impose financial penalties; the standing threat of these remedies is what makes commercial contracts effective even in the cases where the remedies are not invoked. Williamson [1985]'s transaction-cost analysis treats the cost of using courts as one of the components of market enforcement that hierarchy (the firm) is sometimes able to economize on; the analysis assumes the courts are available as the residual enforcement mechanism.

The commercial-law apparatus is not equally available across all cases. Cross-border transactions involve choice-of-law and choice-of-forum complexities that often render litigation prohibitively expensive. Transactions between parties in jurisdictions whose courts do not recognize each other's judgments lack effective remedy. Transactions between informally-constituted parties, parties without legal personality in the relevant jurisdiction, or parties whose identity is hard to establish, are similarly under-served. These under-served cases have historically been managed by intermediaries (merchant houses, factors, escrow agents) or by extralegal arrangements (reputation, repeated interaction). Each of these secondary mechanisms operates in the shadow of the residual state-coercion option, even when the option is not directly available — because the residual option's existence has shaped the social conventions on which the secondary mechanisms rely.

Coercion as a variable. Treating coercion as a *variable* that can in principle be set to zero is, like treating subordination as a variable, a move that the dominant tradition has not historically been able to make. The reason is that no coordination architecture has been available in which bilateral commercial obligations could be enforced at scale without a residual coercive backstop. The bond posted by the parties under prior arrangements was either (a) held by an intermediary who would release it to the cooperative party — in which case the intermediary's discretion was the coercive mechanism, simply relocated; or (b) held by a court — in which case the court's standing capacity was the coercive mechanism, simply made explicit; or (c) absent — in which case the obligation was un-enforced and the parties bore the performance risk. The dominant tradition's analysis was correct on its own terms: under the available coordination architectures, some form of coercive backstop was structurally necessary.

The bonded settlement primitive is a coordination architecture in which the bond is held neither by an intermediary nor by a court, and in which the release is performed by a public algorithm with no discretion. This is the architectural change that makes coercion a variable. The variable can now in principle be set to zero in the bilateral-commerce domain, and the question of what happens to the dominant tradition's analysis when that occurs is the question the present paper addresses.

Remark 2.1 (On reading coercion as a variable). The framing tracks the framing of subordination in the companion partition-conditionality paper. Both moves rely on the fact that political philosophy and political economy have historically treated certain features of social cooperation as structural constants because no architecture admitted their removal. When an architecture does admit their removal, the features become variables, and the analyses that took them as constants relocate accordingly. The relocations are different in each case because the analyses were different, but the underlying move (constant becomes variable; analysis relocates) is the same.

3 What the Primitive Does to Coercion

We now turn from the structural framing to the specific mechanism by which the bonded primitive substitutes for the coercive apparatus in bilateral commerce. The mechanism is established in the companion mechanism paper; we summarize only what is necessary for the present argument.

The substitution rests on two compositions in the kernel. *Asymmetric bonding* (Mechanism 1) supplies the credible threat of loss that makes performance the rational strategy — the substitute for the sovereign's coercive sword in the bilateral case. *Buyer dominance with atomic resolution* (Mechanism 2) supplies the non-discretion that makes the substitute architectural rather than dependent on a trusted enforcer: only the buyer can trigger resolution, and resolution settles every active order together or not at all, with no agent — arbitrator, governance body, oracle, jury — in a position to defer, partial-resolve, or selectively enforce. The structural difference between Hobbesian sovereignty and the bonded primitive is not just that the parties post their own swords; it is that no third party stands in the position from which the sovereign formerly acted.

Pre-commitment instead of retribution. The bond is locked at the moment of consent, from the parties' own resources, by their own signatures. The amount is fixed by the protocol's bonding ratios (twice the contract price for the buyer, twice the cumulative seller value for the seller), and is verifiable by both parties before the commitment is signed. The protocol's resolution mechanic releases the bond to the cooperative party and forfeits it from the defecting party, automatically, with no discretion exercised by any human agent or external authority — the second mechanism above is what guarantees that no such discretion exists. The mechanism is what Elster [1979] analyzed under the heading of *pre-commitment*: a self-binding move at the moment of consent that substitutes for ongoing enforcement during performance.

Pre-commitment differs from retributive coercion in three structural ways. First, it operates at the moment of consent rather than at the moment of breach: the binding occurs when the parties agree, not when one of them tries to deviate. Second, the resource that performs the binding is the party's own, posted voluntarily, rather than the sovereign's standing capacity to apply force. Third, no agent is in the position to apply or withhold the binding after the fact: the protocol executes the release or forfeiture mechanically, without discretion, and without the structural capacity to be lobbied, captured, or selectively enforced.

Hobbes's covenant with self-applied sword. Hobbes's famous remark that "covenants, without the sword, are but words" identifies the structural problem the dominant tradition has spent four centuries solving by erecting and legitimating sovereign coercive apparatuses. The bonded primitive offers a different solution: the covenant arrives with its own sword, applied automatically by the protocol upon breach, with the sword being the bond the breaching party has itself posted. The Hobbesian framing is not refuted; it is satisfied by a different mechanism. The covenant has its sword; the sword is not the sovereign's; the parties supply it themselves at the moment of agreement.

This is a literal reading, not a metaphorical one. The bond is the sword in the precise sense Hobbes intended: a credible threat of loss whose existence makes performance the rational strategy. What changes is who holds and who applies the sword. Under the sovereign's monopoly, the sovereign holds and applies; the parties' obligations are effective because the sovereign stands behind them. Under bonded pre-commitment, the parties themselves hold (in escrow with the protocol) and the protocol applies; the parties' obligations are effective because they have themselves made the breach more costly than the performance.

What this is not: a softer kind of coercion. A reader might object that pre-commitment is just coercion with a friendlier name — the bond is forfeited against the breaching party's preferences, after all, and that is coercive in the ordinary-language sense. The objection elides a structural difference that the political-philosophical literature has long recognized. Nozick [1969] distinguished proposals from threats by reference to the baseline against which compliance is evaluated: a threat alters the baseline in a way that compels compliance, while a proposal offers an alternative to the baseline that the recipient may accept or refuse. The bond posted at consent is a proposal in Nozick's sense: at the moment of bonding, the party's baseline is unchanged; the bond arrives only with the party's signature, and the consequences of forfeiture are baked into the offer. This is the ordinary structure of voluntary commerce. It is categorically different from the standing capacity of a sovereign to apply force against a party who has not consented.

Remark 3.1 (On the limit of the analogy). The pre-commitment / retribution distinction is structural, not ethical. We are not claiming that pre-commitment is morally superior to retributive coercion — that is a normative question on which the substrate takes no position. We are claiming the two are different mechanisms of enforcement, with different agents, different timing, different relations to consent, and therefore different political-philosophical analyses. The substitution of pre-commitment for retribution in a bounded domain therefore changes the political-philosophical analysis applicable in that domain. This is descriptive, not prescriptive.

4 Distance and the Menu of Coercive Alternatives

A natural question is when, in practice, the bonded mechanism is the operative enforcement mechanism and when it is one of several coexisting mechanisms. The answer is governed by what we will call *social distance* between the parties, by which we mean their distance from one another along three dimensions: physical-geographic, jurisdictional-legal, and identity-relational.

Physical-geographic distance. Parties in physical proximity have access to enforcement mechanisms that do not require either bond or sovereign: in-person renegotiation, social pressure, recourse to local intermediaries, and at the limit, direct physical compulsion. The bonded mechanism is available to them but is not the only option. Parties at physical distance lose access to most of these alternatives. Direct compulsion becomes infeasible; in-person renegotiation cannot occur; social pressure operates only insofar as the parties share a community that traverses the distance.

Jurisdictional-legal distance. Parties under a shared sovereign jurisdiction have access to that sovereign's coercive apparatus: courts, contract enforcement, specific performance, damages, criminal sanction for fraud. Parties under distinct sovereigns may or may not have access, depending on the existence of mutual-recognition treaties, the cost of cross-border litigation, and the political-economic relationship between the sovereigns. Parties under sovereigns that do not recognize each other's judgments, or under no sovereign at all, lack the apparatus entirely.

Identity-relational distance. Parties who know each other, who have transacted before, or who share membership in a small community have access to reputational enforcement: the cost of breach includes the loss of future transactions and standing within the community. Parties who are strangers to each other, who have no prior interaction, and who share no community lack this. The reputational mechanism does not extend to first encounters between unknown parties.

The menu argument. At low social distance along all three dimensions, parties have a rich menu of enforcement mechanisms: physical, sovereign-legal, reputational. The bonded mechanism is one option among several and is not necessarily the dominant one. At high social distance along all three dimensions, the alternatives drop away: physical compulsion is infeasible, sovereign apparatus is unavailable or prohibitively expensive, and reputation cannot operate between strangers. The bonded mechanism is then the only enforcement mechanism available, and its existence becomes the difference between transaction and no transaction.

This is not an empirical claim that the bonded mechanism will be adopted at high social distance and ignored at low social distance — adoption is a separate question — but a structural observation about where the mechanism is doing decisive work and where it is one option among others. The decisive cases are the ones in which prior coordination architectures had no answer: cross-border commerce between parties with no shared sovereign, exchange between parties whose identities cannot be reliably established, transactions between artificial agents acting under distributed organizational arrangements, commerce by populations who lack civil-legal personality in any jurisdiction that the counterparty can sue in.

Remark 4.1 (On the physical-proximity intuition). A common intuition holds that the bonded mechanism's incentives are stronger at distance and weaker in proximity. The intuition is correct in its direction but imprecise in its mechanism. The bonded mechanism's incentives do not change with distance — the bonding ratios are constant. What changes with distance is the menu of competing mechanisms. At distance, the menu shrinks to the bonded mechanism alone, making it decisive. In proximity, the menu expands to include alternatives, of which the bonded mechanism is one. The mechanism is not stronger at distance; it is exclusive at distance and one-of-several in proximity.

5 The Boundary: What Shrinks, What Stays

The bonded mechanism's domain of substitution is bounded. Within the boundary, it substitutes for sovereign-backed coercion; the coercion is no longer being applied because the obligation is self-enforcing. Outside the boundary, the bonded mechanism is structurally unable to substitute, and sovereign-backed coercion continues to perform functions that the bond cannot. This section identifies what is on each side of the boundary, and why.

Inside the boundary: priceable bilateral commercial obligations. The bonded mechanism substitutes for coercion in obligations that are (a) priceable in a unit the protocol denominates bonds in, (b) bilateral or composable into bilateral edges via the mechanism's progressive collateralization, (c) voluntary at the moment of bonding, and (d) capable of being specified in schemata that the parties accept at consent. The bulk of contemporary commercial activity falls under these conditions: sale of goods, provision of services, delivery, escrow, payment, attestation against measurable standards. For these, the bonded mechanism removes the requirement for an external coercive backstop.

The substitution has cascading effects on the regulatory apparatus that historically performed coordination functions adjacent to direct enforcement. Antitrust regimes assume the existence of intermediaries with market power to constrain; where intermediation is structurally unnecessary, the intermediary-targeting regime has less to operate on. Consumer-protection regimes assume an information asymmetry between merchant and consumer that schema-validated attestations substantially reduce. Cross-border commercial-dispute regimes assume the necessity of choosing between competing sovereign coercive apparatuses; where the obligation is self-enforcing, the choice is moot. The shrinkage is real but is itself bounded: each of these regimes addresses other concerns (systemic risk, fraud, market manipulation) that survive the substrate change.

Outside the boundary: non-priceable, non-bilateral, non-voluntary harms. The bonded mechanism cannot substitute for sovereign-backed coercion in cases where the obligation is non-priceable (violence against persons cannot be made commensurable with a sum of money in a way that captures what the violation involves), non-bilateral (third parties harmed by an externality are not parties to the bond), or non-voluntary (the harming party did not consent to a bond at the moment of harm). These include criminal harm against persons, large-scale environmental externality, public-goods provision under free-rider conditions, status questions (citizenship, legal personality, identity), fundamental rights adjudication, and the standing capacity to defend the territory against external coercion.

For each of these, the sovereign's coercive apparatus continues to do work that the bonded mechanism is structurally unable to do. The reason is not that the bonded mechanism is inefficient in these domains but that the conditions for its operation are not present: there is no consent at the moment of harm, no bilateral counterparty to bond against, no priceable specification of the obligation. Substituting bond for sovereign in these domains would require the substitution to itself be sovereign-backed, which is not a substitution.

The boundary itself. The boundary between the two domains is not a moral or political choice but a structural one, given by the conditions under which bonded pre-commitment is operable. The boundary will shift over time as schema design improves and as bondable composition extends to obligations that are not currently priceable in practice. But the structural limits — non-consent at the moment of harm, externality on non-parties, non-priceability of certain violations — are categorical and do not yield to engineering improvement. They mark the part of the cooperative landscape where sovereign-backed coercion remains load-bearing.

What this implies for the apparatus. The state's coercive apparatus is not made unnecessary by the substrate change. It is bounded, in the precise sense that some of its prior functions (commercial enforcement of bilateral obligations) are no longer required for the obligations to be honored, while other functions (criminal, externality, public goods, status, rights, defense) continue to require it. The shrinkage is in the part of the apparatus's work that was addressing the coordination problem; the part addressing the not-coordination problems remains.

Remark 5.1 (On contested boundary cases). Some categories sit on the boundary and admit dispute about which side they fall on. Securities regulation is one example: some of its function is intermediary-targeting (and shrinks with the substrate change), some is externality-pricing (systemic risk to non-parties, which does not shrink), some is status-based (defining qualified investor categories, which does not shrink). Disentangling these is empirical work the present paper does not undertake. We note only that the boundary is real, that it bounds the political-philosophical question, and that contested cases at the boundary are adjudicable on structural grounds rather than political ones.

6 The Democratic-Republican Apparatus: Bounded Domain, Not Reformed

We now turn to what the substrate change implies for the democratic-republican apparatus through which contemporary states exercise their coercive monopoly. The implication is narrower than may be tempting to claim, and the framing matters.

The levers of power and their abuses. The standard analysis identifies four broad levers of power in democratic-republican constitutional systems: legislation (rule-making), executive enforcement, judicial adjudication, and the franchise (the citizenry's authorization of the previous three). Each is subject to characteristic abuses: regulatory capture in legislation (Stigler 1971), executive overreach and emergency powers (Schmitt 1922), judicial drift, gerrymandering and voter suppression in the franchise, and across-lever phenomena like agency capture and lobbying (Olson 1965). These abuses are real, are extensively documented, and have generated centuries of political-philosophical and constitutional reflection on how to constrain them.

What the substrate change does not do. The substrate change does not correct the abuses listed above. Regulatory capture continues to operate in the domains where regulation continues to operate; executive overreach continues to be possible in the domains where the executive continues to exercise authority; judicial drift continues in the domains where courts continue to adjudicate. The substrate change does not provide a constitutional remedy for any of these. It is not a proposal for democratic reform.

What the substrate change does do. The substrate change shrinks the *domain* over which the levers of power operate. Where bilateral commercial obligations become self-enforcing, the legislative apparatus that has historically generated rules for their regulation has less to generate rules about. Where commercial disputes resolve atomically by protocol, the judicial apparatus that has historically adjudicated them has less to adjudicate. Where coordination occurs by bond, the executive apparatus that has historically enforced commercial obligations has less to enforce. The abuses that operate on these levers are not corrected; the levers' *reach* is narrowed in the part of social cooperation that becomes self-enforcing.

This is subtraction, not reform. The political-philosophical work of constraining the levers in their remaining domains continues. The work is, however, more bounded than it was: the domain over which legitimate coercion must be authorized, exercised, and constrained is smaller, because a substantial part of what historically required authorization, exercise, and constraint is now operating without coercion in the relevant sense.

Pettit's freedom-as-non-domination, and where it lands here. Pettit [1997] reframes liberty in a way the present argument can use directly. The classical liberal reading takes liberty to be the absence of *actual* interference; an agent is free to the extent that no one is currently obstructing their actions. Pettit's republican reading takes liberty to be the absence of *capacity* for arbitrary interference; an agent is free to the extent that no one is in a position to interfere arbitrarily, even if no one is currently doing so. The republican reading is a stronger condition: a slave with a kind master is not free in the republican sense, because the master retains the capacity to interfere arbitrarily even when not exercising it.

The bonded primitive's non-discretion is a freedom-as-non-domination property. At the moment of resolution, no third party is in a position to interfere arbitrarily with the outcome — not the buyer (whose dominance is bounded by the conservation law and the atomic-resolution constraint), not the seller, not a manager, not a foundation, not a court (the court can adjudicate on the evidence but cannot reach into the bonded escrow). The participants in a bonded process enjoy republican-style freedom within the process's scope: not because no one chooses to interfere, but because the architecture has eliminated the position from which arbitrary interference could be exercised. The connection should not be over-stated — republican freedom is a property of political order broadly, and the bonded primitive is a settlement architecture in a specific domain — but the structural alignment is direct, and the primitive supplies within its domain what republican theory has historically required political institutions to supply across domains.

What about democracy itself. A democratic-republican order has multiple justifications, only some of which depend on the order's coordinative function in commerce. The order's role in legitimating coercion (Locke's consent of the governed), in protecting rights against arbitrary authority (Madisonian checks), in providing for collective self-determination (the republican tradition from Cicero through Pettit [1997, 2012]), and in adjudicating among competing visions of the good (the deliberative tradition from Mill through Habermas), are each independent of the order's role in regulating commerce. None of these is affected by the substrate change. The order's coordinative function in commerce, however, is one of its historically major functions, and that function shrinks. A democratic-republican order is not made unnecessary by the substrate change; it is freed, in the bounded domain of self-enforcing commerce, from one of the responsibilities that has historically occupied it.

Remark 6.1 (On the Network State and adjacent proposals). A separate question, on which we take no position here, is whether the substrate change implies that some functions of the democratic-republican order can or should be replaced by network-organized arrangements (Srinivasan 2022). That question requires its own treatment and is the subject of a forthcoming companion paper. The present paper notes only that the substrate change does not by itself entail any particular network-organizational form, and that conflating the substrate change with a specific political proposal is the category error this paper has been at pains to avoid.

7 What This Paper Is Not

The argument is liable to several misreadings; we address them explicitly.

Not anarchism. The paper does not argue that the state is unnecessary, that the sovereign's monopoly on legitimate violence should be dissolved, or that the coercive apparatus should be abolished. The state's load-bearing functions outside the bilateral-commerce domain are preserved by the analysis. A reader who concludes that the paper endorses anarchism has read shrinkage in one domain as abolition across all domains.

Not libertarianism. The paper does not argue that the state *should* be smaller, that voluntary exchange is morally superior to authoritative coordination, or that the night-watchman state of Nozick [1974] is the right institutional configuration. The paper makes no normative claim about whether the shrinkage is good or bad, only the structural claim that it occurs in a bounded domain. A reader who concludes that the paper endorses libertarianism has read structural observation as normative recommendation.

It is worth distinguishing the present argument from the adjacent libertarian-anarchist program. Nozick [1974], Friedman [1973], and the broader anarcho-capitalist tradition argue that the state's enforcement function can in principle be replaced by competing private protection agencies, and that the resulting arrangement would be morally and practically superior to the state. The structural claim in those programs is *unbounded substitution*: every coercive function the state performs can in principle be performed by some private arrangement, including criminal law, defense, and adjudication of disputes between strangers. Our claim is the structurally weaker *bounded substitution*: a specific class of bilateral commercial obligations becomes self-enforcing under the bonded primitive, and *outside that class* the conventional sovereignty apparatus remains in place and is not analyzed here. The substitution is bounded because the primitive's non-discretion is purchased by accepting the constraint that the substituted function be specifiable in advance; the criminal-law function and the broader adjudicative functions of the state are not so specifiable, and the bonded primitive does not pretend to substitute for them. A reader sympathetic to the libertarian-anarchist program should note that the present argument supplies neither the unbounded substitution they may hope for nor a refutation of the desirability of such a program; it supplies a bounded substitution in one domain and is silent on the others.

Not statism. Symmetrically, the paper does not argue that the state's remaining functions are well-performed, well-justified, or that the apparatus that performs them is in good order. The paper is silent on these questions. The fact that the bonded mechanism does not substitute for the state's criminal-law function does not entail that the state's criminal-law function is being performed correctly. That is a separate inquiry on which the substrate takes no position.

Hobbes and Schmitt: how the substitution does and does not engage them. Hobbes and Schmitt are the two thinkers whose challenges to the present argument are most often raised, and they require different responses.

Hobbes's structural observation, that covenants without swords are but words, is satisfied — in the bilateral-commerce domain and only there — by a different sword: the bond the parties have themselves posted, released or forfeit by an algorithm whose execution they have themselves authorized. The Hobbesian challenge is met on its own terms in the domain where the substitution applies, and is not addressed in the domain where the substitution does not apply. We do not argue that the substitution generalizes outside bilateral commerce.

Schmitt's challenge is sharper and more interesting, and we engage it directly rather than dismiss it. Schmitt [1922] argues that any rule-system requires an extralegal decision on the exception, and that the sovereign is whoever decides whether a given case is or is not a normal application of the rules. The bonded primitive, taken as a rule-system, would on Schmitt's account require some sovereign to decide when it does not apply. Our response is that the primitive is not a rule-system in Schmitt's sense; it is a settlement primitive that runs in the domain where its rules apply and is silent in the domain where they do not. The Schmittian decision on the exception — whether a particular bonded commitment was procured under duress, whether a particular outcome violates public policy, whether a particular party lacks capacity — is a decision the primitive does not make and does not pretend to make. It is made, where it is made at all, by external legal forums applying their own constitutional and doctrinal apparatus to the bonded commitment as evidentiary input (the companion evidence-and-law paper develops this in detail). The exception is not absorbed by the primitive; it is left where it has always been, in the forums whose authority to decide on it derives from elsewhere.

The Schmittian challenge therefore lands, but lands at a different level than its formulators may have intended. It is correct that the primitive cannot decide its own exceptions; that is precisely why the exception remains the province of external forums, and why this paper is careful to bound its substitution claim to bilateral commerce in normal application. The non-discretion of the primitive within its domain, and its silence outside its domain, are two faces of the same design property: the primitive has no authority to extend itself, and no authority to retract itself either. That property is what makes coexistence with conventional sovereignty architectural rather than negotiated.

Not a constitutional proposal. The paper does not propose constitutional reforms, alternative governance structures, replacement of legislative or judicial institutions, or any other prescriptive change to existing political arrangements. The substrate change occurs at a layer beneath constitutional choice, in the same way that the electrical substrate underlying contemporary economies occurs at a layer beneath constitutional choice. The constitutional question of how to organize collective coercion in the remaining domain is unaffected by the substrate change at that layer; what is affected is the domain over which the question must be answered.

Not a prediction. Finally, the paper does not predict that the state will wither, that bilateral commerce will fully migrate to the bonded substrate, or that the political consequences of such migration will manifest in any particular timeframe or distribution. The argument is conditional: under the supposition that the bonded mechanism substitutes for sovereign-backed coercion in its operable domain, the political-philosophical analysis of coercion relocates in the manner described. Whether the supposition obtains in practice is the empirical question of adoption, which the present paper does not address.

8 Conclusion

This paper has argued that coercion is a substrate variable analogous to subordination and coordination rent: structurally load-bearing in the dominant tradition's account of bilateral commercial enforcement, and substituted by pre-commitment when the bonded primitive is the operable coordination architecture. The substitution holds within a bounded domain (priceable, bilateral, voluntary, schematizable obligations) and does not extend beyond it (criminal harm, externality, public goods, status, rights, defense). Within the domain, the political-philosophical question of legitimate coercion becomes irrelevant rather than corrected; outside the domain, the question retains its full urgency, now operating over a bounded rather than an unbounded scope.

The substantive observation is that the philosophical analysis of coercion that has organized political thought from Hobbes through Weber, Hart, and the contemporary literature has been implicitly ranging over the entirety of social cooperation because no architecture admitted bilateral commercial enforcement without coercive backstop. With such an architecture available, the analysis becomes bounded. This is not a refutation of the prior tradition — the analysis remains correct on its own terms in the remaining domain — but a relocation of where the question of legitimate coercion must be answered. The state retains its load-bearing functions; the question of how it should exercise them retains its political-philosophical urgency; what changes is the perimeter within which the question operates.

We close with two observations of method, mirroring the closings of the companion political-economy papers in the corpus. First, the kernel is ideologically agnostic; the graph is the politics. What the present paper adds to that formulation is that the political-philosophical analysis of coercion, like the political-economic analysis of subordination, becomes a graph-tier question over a bounded substrate domain rather than a substrate-tier inevitability over an unbounded one. The politics is at the graph in a stronger sense: it is freed, in the bounded domain, from a structural premise (coercive backstop required) that prior architectures forced upon it. Second, the relocation described here is recursive. Future substrates may relocate further variables from architecture to choice, and the boundary between bonded substitution and sovereign retention may itself be re-examined as schema design extends what counts as priceable. There is no terminal architecture; there is only the next architecture whose variables we have not yet learned to see as variables.

Acknowledgments

The argument of this paper began as an intuition — that the mechanism's effectiveness varied inversely with the availability of coercive alternatives, and that the dominant tradition's analysis of law as text-plus-coercion required revision in the substituting domain — voiced in conversation. The intuition was correct in direction but imprecise in mechanism, and this paper is the unpacking of the more precise structural observation to which it points. I thank the interlocutor whose intuition prompted the unpacking, and the readers of earlier drafts who pressed me to scope-bound the argument carefully and to resist the triumphalist framings that the topic invites.

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